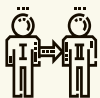


The background is a deep space scene. It features a dark, star-filled sky. There are wispy, glowing green and blue nebulae scattered across the frame. In the upper right quadrant, a large, dark, spherical planet or moon is visible, partially obscured by the main title text.

ROVER RESCUE

PROMPT 2

Charli Dolan, Daniela Marcano, Rio Johnson, Jim Guadagnoli



OUR TEAM



DANIELA
MARCANO



RIO
JOHNSON



JIM
GUADAGNOLI



CHARLI
DOLAN

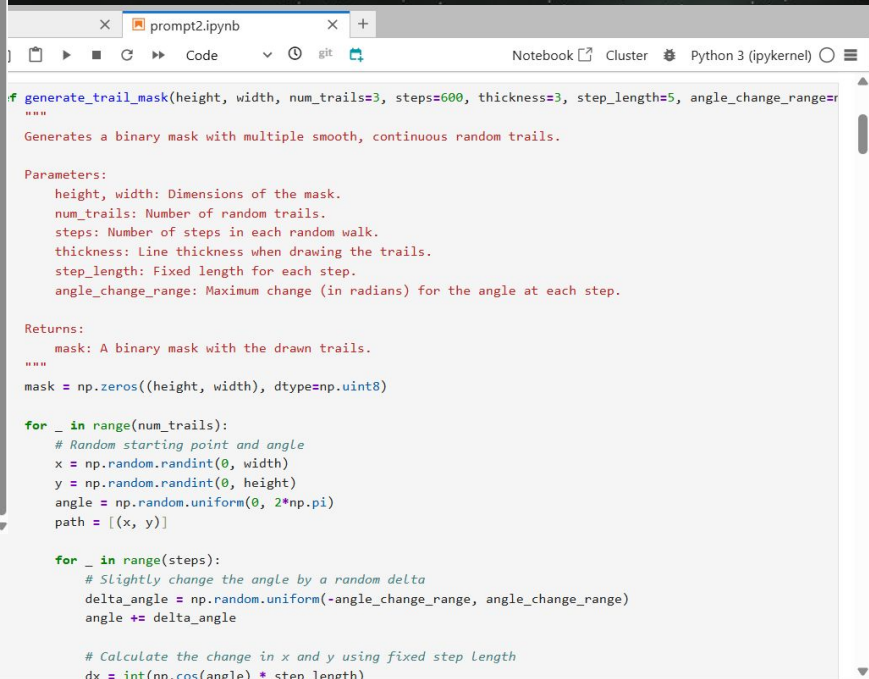
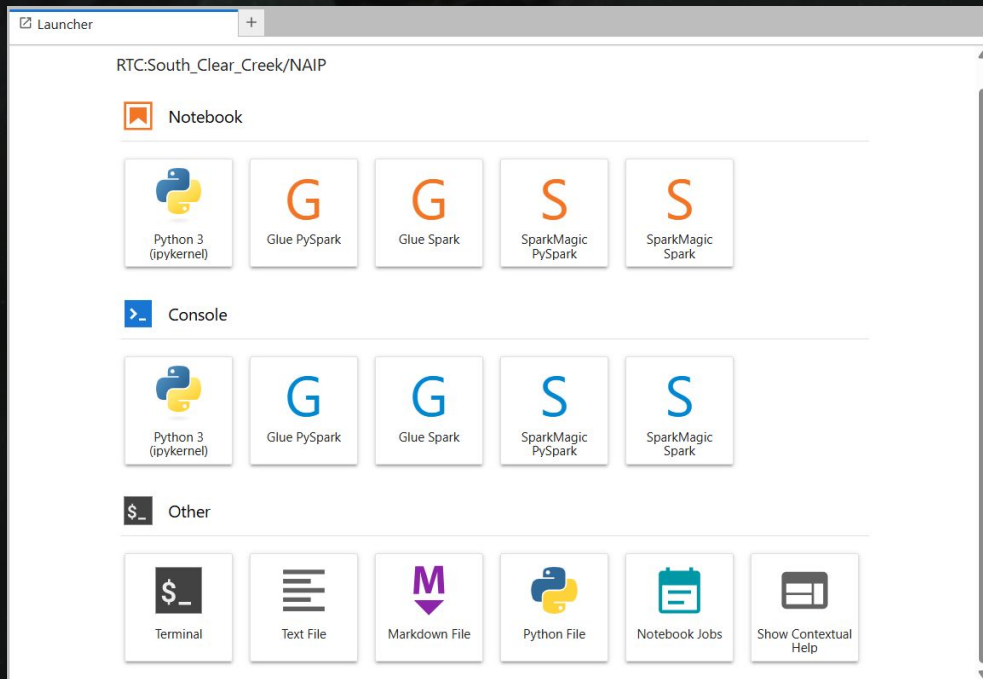




01

INITIAL APPROACH

Understanding the prompt



JUPYTER IDE

Visual Studio Code

Editing evolved

Start

- New File...
- Open File...
- Open Folder...
- Connect to...

Recent

Hackathon C:\Users\soren\Downloads

Walkthroughs

- Get Started with VS Code
Customize your editor, I
- Get Started with Python
- Learn the Fundamentals

starter_code_nasa > starter_code_nasa > __pycache__ > Data.py > TiffCenterCropper > load_and_crop_tiff

```
1 import cv2
2 import numpy as np
3 import pygame
4 import tifffile
5
6 class TiffCenterCropper:
7     def __init__(self, file_path, crop_size=8000):
8         """
9         Initialize the TiffCenterCropper.
10
11         Parameters:
12         - file_path: path to the TIFF file
13         - crop_size: size of the square crop (default 8000x8000 pixels)
14         """
15         self.file_path = file_path
16         self.crop_size = crop_size
17         self.cropped_image = None
18         self.load_and_crop_tiff()
19
20     def load_and_crop_tiff(self):
21         """
22         Load the TIFF file and crop the center portion.
23         """
```

VSCODE IDE

CHAT GPT

What can I help with?

Ask anything

+ Search Reason ...

🎤

Services [Show more ▶](#)

 **Amazon Bedrock** ☆
The easiest way to build and scale generative AI applications with foundation models...

Top features

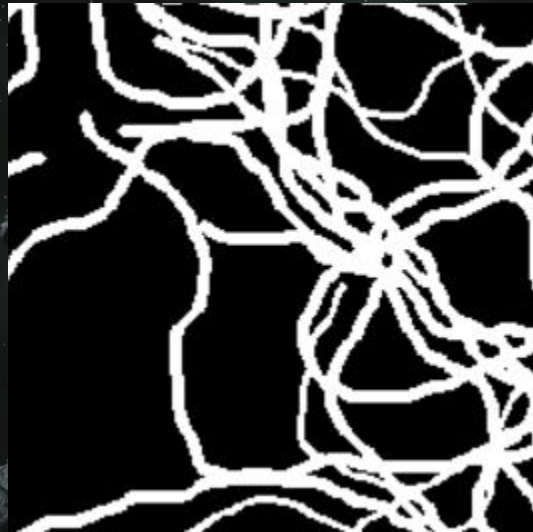
[Agents](#) [Guardrails](#) [Knowledge Bases](#) [Prompt Management](#) [Flows](#)

BEDROCK

LLM's

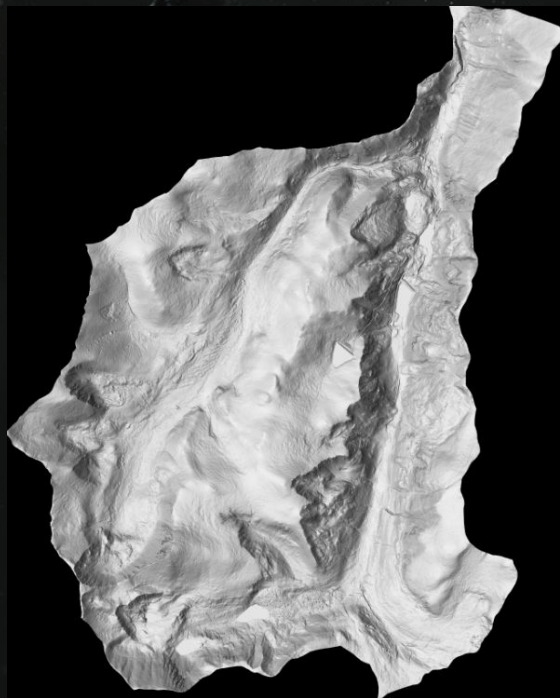
```
import pandas as pd
from sklearn.preprocessing import StandardScaler, LabelEncoder
from torchvision import transforms
from PIL import Image
import torch
from torch.utils.data import Dataset, DataLoader
import os
import cv2
import torchvision.transforms as transforms
from rasterio.plot import rasterio
from rasterio.plot import show
import matplotlib.pyplot as plt
import networkx as nx
import networkx as nx
import matplotlib.pyplot as plt
import numpy as np
import random
```

LIBRARIES



RANDOM MAP GENERATION

GEOGRAPHICAL .TIF



MASK .TIF



MASK .NPY



MAPS

The background of the slide is a dark, space-themed image. It features a large, reddish-brown planet with visible surface details, partially obscured by a dense field of grey, irregularly shaped asteroids of various sizes. The overall lighting is dim, with some highlights on the planet's surface and the edges of the asteroids.

02

PROBLEM SOLVING

Different approaches taken, their drawbacks, and success

CHALLENGES

Problems

- We struggled at first, jumping between resources and getting disorganized.
- Jupyter Labs added to the confusion.
- We accidentally started Prompt 1 instead of Prompt 2 and had to restart.
- The 'South_Clear_Creek_Roads_Mask' file exceeded Pygame's capacity, causing data limitations.
- Data corruption was a major obstacle that impacted our ability to work with NAIP files

Solutions

- We regrouped with determination for Prompt 2 and space rescue.
- Our strong team bond kept us motivated.
- Google Docs helped us stay organized.
- Staff support guided our progress.
- We created Pygame-compatible datasets to bypass data limits.
- We refined our AI for better pathfinding.
- We resized the given data to fit Pygame's constraints.



03

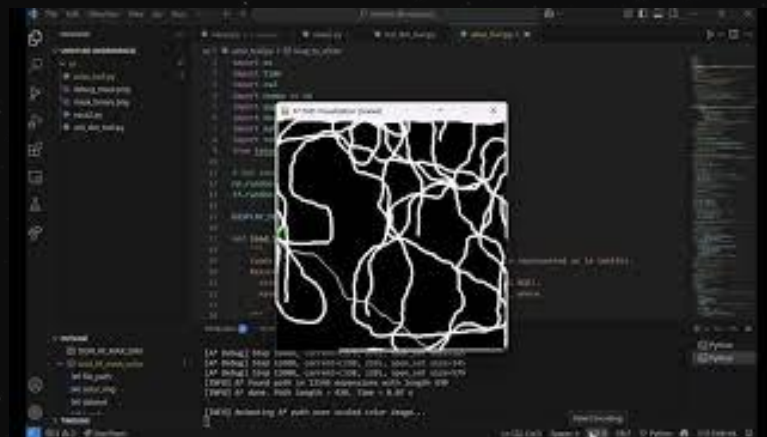
FINAL PRODUCT

A working, path finding rover

ROVER

Random Generation Map

<https://www.youtube.com/watch?v=52mOLdLtZnI>



Geographical Mask Map

https://www.youtube.com/watch?v=TD00_6Bunf4





04

REFLECTION

Experience that we gained, what we would do differently

STRENGTHS

TEAMWORK

- Prioritized clear communication and a collaborative environment
- Combined varying strengths to enhance problem-solving and creativity

PERSEVERANCE

- Remained committed to our end goal despite numerous setbacks and false starts
- Continued to refocus our efforts and explored unique solutions to push forward

ADAPTABILITY

- Utilized available resources to effectively tackle technical issues
- Sought innovative and original approaches to address challenges

IMPROVEMENTS

WHAT WOULD WE DO DIFFERENTLY?

- Stronger initial planning
- Focused research
- More robust testing

FUTURE PLANS

- Enhancing AI training to optimize dynamic pathfinding
- Improving UI visualization quality and usability

THANKS!

Special thanks to the AWS team, Nasa team, USDA team and the dev net club for making this hackathon possible!

CREDITS: This presentation template was created by **Slidesgo**, including icons by **Flaticon**, and infographics & images by **Freepik**